



Machine Learning–Based Traffic Prediction for Peering Selection in ISP Networks

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The work was commissioned by Elisa Oyj, a Finland-based telecommunications company. The thesis aims to improve Elisa's utilization of private peering vs. public peering or transit providers to reduce networking costs.

The goal for this thesis was to develop a program which uses machine learning to examine past traffic patterns in order to predict future traffic amounts to enable better utilization of investment and proactive building of connectivity.

The research method used was a methods approach that combined quantitative analysis, qualitative insights and applied research.

To understand the topic, the thesis researches the structure of the Internet and how traffic is passed between different parties and the costs associated with these methods. The following research questions are answered: Can we use legacy data to predict future trends? What factors do we need to consider to make good predictions?

The defined research questions were answered in the literature review.

The thesis reaches the set goal of creating a program which utilizes different machine learning models to predict future traffic amounts between Elisa and possible peering partners.

Keywords: Private Peering, BGP, Transit, Machine learning

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1 Introduction

This thesis was done for Elisa Oyj. The objective of this development work is to create a tool which uses legacy traffic data to identify positive trends in traffic amounts between the ISP and possible peering partners. It will also align physical locations where both ISP and the peering partner are both present.

ISP peering refers to the practice where Internet Service Providers (ISPs) connect and exchange traffic directly with one another. This allows data to flow between their networks without needing to go through a third-party network. Peering can be either settlement-free, where both ISPs mutually agree to share traffic without charges, or paid peering, where one ISP compensates another for carrying its traffic.

The benefits of peering include reduced latency, improved network performance, lower costs for data transmission, and enhanced reliability. ISPs often establish peering relationships at Internet exchange points (IXPs) or common data centers to facilitate these direct connections. (AMS-IX)

Currently the selection is done by human evaluating past traffic data and using intuition to determine whether the possible partner is suitable for peering or not. There is no comprehensive tool available to evaluate if peering would be feasible before doing any evaluation of the traffic amounts by hand, nor is there fact-based predictive information available to help start the peering process before we hit the key metrics.

By using Machine Learning to analyze the historical traffic data we gain knowledge on the trend and can evaluate if the traffic is enough to justify starting the peering process. Even better outcome is if we can see that there is high probability that the traffic amount will be reached soon, this would give the ISP better time to invest in hardware and start the peering process with the partner even before the appropriate metrics are reached.

In this thesis the following research questions are answered: Can we use legacy data to predict future trends? What factors do we need to consider to make good predictions?

2 Background

2.1 Public Peering, Private Peering and Transit

Public peering refers to the practice of two networks (typically Internet Service Providers, ISPs, or large networks) connecting to each other at an Internet Exchange Point (IXP) or similar public facility. (IX-F 2025)

Typically public peering is done at a physical location where both networks connect to a shared physical infrastructure (usually at an IXP), and they peer with each other over this shared platform. Usually IXPs are open and shared, multiple networks are interconnected, allowing many peers to exchange traffic with one another. This means that not only do two networks exchange traffic directly, but they can also potentially exchange traffic with any other network connected to the IXP. As the infrastructure is shared, costs for the exchanged data is lower as no dedicated private connections are used. (ThousandEyes 2025) With shared infrastructure the downside is limited control to how the traffic is passed between parties and the IXP itself might have some requirements in place also.

Example: A small regional ISP might connect to a major backbone provider via a public IXP, where they can also connect with other smaller ISPs or content providers (like Google or Facebook).

Private peering refers to the direct connection between two networks without using a public exchange point. These networks establish a private, dedicated link between each other, typically for the purpose of exchanging large volumes of traffic or improving performance as latency, throughput and reliability. (Hedlund 2008)

Private peering involves a dedicated, often fibre-optic, link between the two networks. This avoids the congestion or shared nature of public peering. With private peering, both parties manage the quality of service and routing policies, which can result in better performance, lower latency, and higher security. Large ISPs, content providers, or enterprise networks often prefer private peering because they can manage their traffic more directly, and they may need a lot of bandwidth or special routing conditions that public peering doesn't offer. Private peering tends to be more expensive than public peering because it requires dedicated infrastructure, and typically both parties must invest in the connection, but as the traffic volumes grow, private peering does become less expensive than public peering, or purchasing the traffic volumes from a transit provider.

Example: A content provider like Google will very likely have a private peering arrangement with a major ISP to ensure that traffic between them is fast, reliable, and cost-efficient. This would allow them to bypass public exchange points and handle large volumes of traffic directly.

Transit is a broader term that refers to the process of one network paying another network (often a larger backbone provider) to route traffic between them and the rest of the internet. In a transit arrangement, one network (typically a smaller ISP or enterprise) purchases access to the internet through a larger provider. (Wikipedia 2025)

When a network purchases transit from a larger network, it allows that network to reach all destinations on the internet (including networks that it doesn't have direct peering with). Transit is a paid service, with the smaller network (often called a "customer") paying the larger network (called a "provider") for access to the internet and to other networks. Through transit, a network doesn't need to establish direct peering with all other networks on the internet. Instead, they can route their traffic through their transit provider, which has a large, global backbone and extensive peering relationships. Transit is typically more expensive than peering because the customer is paying for internet access across the entire internet, and the provider is responsible for managing and maintaining the global routing of that traffic.

In figure1, A smaller regional ISP (ISP #1) might purchase transit from a larger national ISP (such as AT&T, Level 3, or CenturyLink) to gain access to the internet, rather than trying to establish peering relationships with every network it needs to reach. Larger ISP, (ISP #4) has chosen to peer with different parties via IXP and private, direct connections. While the

Transit ISP uses its large global presence to connect with all the ISPs and carry their traffic for a fee.

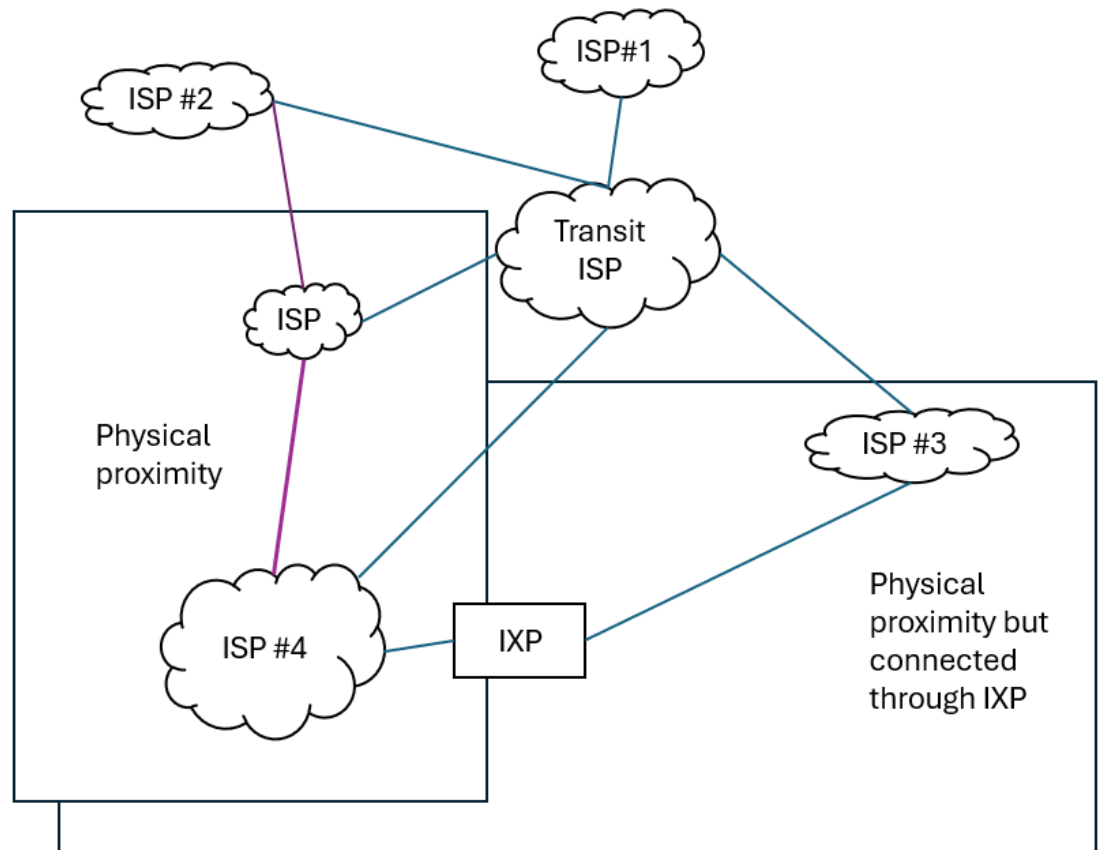


Figure 1. Peering and Transit

2.2 Point Of Presence

Point Of Presence, POP, refers to the location where the ISP has their equipment installed or has direct connections available to. In many cases operators prefer operator independent colocation providers outside their own network area, the major global carrier neutral colocation providers are currently Equinix and Digital Realty, large operators as NTT and Telehouse, which is owned by Japanese operator KDDI, are also large global colocation service providers. (COLO)

2.3 Public Peering

Internet Exchange, or IX, is a local hub where ISP, or a corporation, can connect with others via shared network for a fee. Usually peering in these locations is done without any monetary action between the peering parties, but there are exceptions to this if the network sizes are not equal or the other party is not willing to peer without leveraging some of the cost to the other party (usually due to asymmetry in the traffic). Most Internet Exchanges also publicly list their members, but the most referred source is PeeringDB as it collects all sources to one database. PeeringDB is currently the dominant source of information globally.

2.4 Overview of networking concepts relevant to peering (e.g., BGP, latency, bandwidth).

2.4.1 Border Gateway Protocol (BGP)

The Border Gateway Protocol (BGP) is the foundational routing protocol used to exchange routing information between autonomous systems (ASes) on the internet. (RFC 4271) Each ISP operates its own AS, and BGP enables them to advertise which IP address blocks they can reach.

In peering arrangements, BGP is used to establish and maintain routing policies between partners. BGP configurations determine which routes are preferred, how traffic flows, and how failover and redundancy are handled.

For example, when two ISPs engage in private peering, they each use BGP to advertise only the routes that are mutually agreed upon, optimizing for performance and avoiding transit costs.

2.4.2 Latency

Latency measures the time delay it takes for data to travel from the source to the destination. In networking, it is usually measured in milliseconds (ms). Smaller delay, or latency, is critical for real-time applications such as voice calls, video conferencing and gaming. Latency also has a major impact on TCP traffic throughput, it is a big driver on peering location selection, though many newer real-time applications rely on UDP protocol to overcome TCP latency problems to ensure throughput.

In peering decisions, latency is a significant factor. Direct peering at closer geographical locations (e.g., via Internet Exchange Points or private links) significantly reduces latency compared to routing through a third-party transit provider.

2.4.3 Bandwidth and Throughput

Bandwidth is the maximum capacity of a connection — how much data can be transmitted per second, typically measured in Mbps or Gbps.

Throughput, on the other hand, is the actual amount of data successfully transferred over the network during a given time period.

In peering relationships, high bandwidth connections are essential to support large volumes of data, especially between content-heavy networks (e.g., ISPs and streaming providers).

A key task in ML-based peering partner selection is identifying whether historical throughput trends between ISPs suggest the need for a higher-bandwidth peering link.

2.5 Traditional methods and criteria used by ISPs

2.5.1 Traffic Analysis

ISPs often monitor historical traffic volumes and growth trends between their network and others. If the traffic exchange with a particular network consistently exceeds a threshold, it may justify a peering relationship. This analysis is usually performed manually or via basic traffic monitoring tools, relying on thresholds and heuristics rather than predictive models.

2.5.2 Geographical Proximity, Physical Access and Policies

Network Reachability: The ability of a potential peer to provide access to a large number of networks or end-users is a significant factor. ISPs look for peers that can enhance their network reachability and redundancy. (ThousandEyes 2025)

Each ISP defines its own peering policy, which sets criteria such as:

- Minimum traffic volume (e.g., 1 Gbps sustained traffic)

- Balanced traffic ratios (inbound vs. outbound)

- Technical capabilities (e.g., BGP support, redundancy)

- Network quality (e.g., low packet loss, acceptable jitter)

These policies are often published in PeeringDB, a public database where networks advertise their peering requirements. (ThousandEyes 2025)

2.5.3 Business Relationships and Reciprocity

In many cases, peering is established based on existing business relationships or informal agreements between network operators. Reciprocity – or mutual benefit – is a driving principle; networks aim to peer where the relationship is expected to be balanced and equitable. This dynamic aligns with game theory, particularly the notion of repeated games, where sustained cooperation emerges when mutual benefits outweigh short-term incentives to defect. (Axelrod 1984) In order to build relations with each other, ISPs participate in regional and global peering forums such as:

- NANOG (North American Network Operators' Group)

- RIPE (Réseaux IP Européens)

- Euro-IX and others

These forums facilitate face-to-face meetings, relationship-building, and peering negotiations. As many of the hidden, upcoming traffic is not public knowledge, but rather something which is only discussed face-to-face in confidence, attending these forums is vital for successful peering operations.

2.6 Limitations of Current Approaches

Despite being effective in many cases, these traditional methods suffer from several limitations such as:

- Manual evaluation is time-consuming and error-prone

- Heuristics fail to capture future traffic trends

- Reactive decision-making may delay peering setups, missing optimization opportunities

- No standardized tool exists for predictive, data-driven partner evaluation

These limitations create a strong use case for the application of Machine Learning, which can automate and enhance the peering selection process using historical data and statistical inference.

3 Research methodology

This chapter describes the need to develop methods for enhancing network performance, enhancing resiliency and lowering operational costs by predicting possible peering partners ahead of the actual moment when statistics show it is viable.

Elisa is the largest ISP in Finland, having a large corporate and consumer customer base with varying needs of connectivity within internet. Such broad needs come with challenges when selecting partners to do private peering connections with. The goal of this thesis work is to create a tool which can help shorten the timeframe which it takes to build a new private peering connection with a new partner.

3.1 Research approach and data collection

The research approach is mainly quantitative as all the source data is derived from current running systems within Elisa. As the source material for the program created during this thesis work is proprietary to Elisa and can be considered a trade secret, the source and data can not be disclosed. What can be said about the data, is that it gives good understanding of traffic amounts between different network entities at a given time, but it does not contain any

specific details about customers nor the information they are transmitting or receiving at that time.

As this thesis work is considered as a proof of concept, the data collection and granularity of the data is limited in sample size and length of period. Even with these limitations it does provide enough evidence to justify consideration of future development.

4 Machine Learning Fundamentals

4.1 Overview of Machine Learning

Machine Learning (ML) is a subfield of artificial intelligence (AI) concerned with developing algorithms that allow computers to learn and make predictions or decisions based on data. Machine learning systems identify patterns and infer rules directly from the data, making them well-suited to solving problems that are difficult or impractical to define manually.

4.2 ML and its categories

4.2.1 Supervised, unsupervised and semi-supervised Learning

In supervised learning, the algorithm is trained on a dataset which is labeled, meaning that each training example is paired with an output label. Common tasks include:

Classification: Predicting discrete labels (e.g., phishing detection).

Regression: Predicting continuous values (e.g., gold prices).

Unsupervised learning deals with unlabeled data. The goal is to find hidden patterns or structures in large amount of data. Examples include:

Clustering: Grouping similar data points (e.g., customer segmentation).

Dimensionality Reduction: Simplifying data while preserving structure (e.g., PCA).

Semi-Supervised Learning Combines a large amount of unlabeled data with a small amount of labeled data, improving learning accuracy as labeling is expensive or time-consuming for large data sets.

4.2.2 Reinforcement Learning

An agent learns to make decisions by interacting with an environment, each decision is being graded by positive or negative points. It is widely used in robotics, game AI, and autonomous systems.

4.3 Specific techniques applicable to the problem (e.g., decision trees, neural networks, clustering).

To evaluate potential private peering candidates based on ISP traffic data, several forecasting and machine learning techniques can be used, each offering different advantages. Moving Average is one of the simplest methods, used to smooth out short-term traffic fluctuations and highlight longer-term trends. This can help identify AS peers with consistently high traffic volumes. ARIMA (AutoRegressive Integrated Moving Average) is a traditional time series model that works well when traffic patterns are stable and show linear trends or seasonality. Prophet, developed by Facebook, is another time series tool that is more flexible and can handle sudden changes or multiple seasonal patterns in the data, making it useful for real-world ISP traffic. LSTM (Long Short-Term Memory) is a neural network that can learn from sequences of data and is especially good at capturing complex, non-linear trends over time. This makes LSTM a powerful option for forecasting traffic that doesn't follow simple patterns. In addition to these time-based models, XGBoost is a machine learning algorithm that can analyze a combination of different features—such as traffic volume, growth rate, and variability—to help rank or classify AS peers based on their potential for private peering. Utilizing a mix of these techniques, ISPs can better understand traffic behavior and make informed decisions about which AS peers are suitable for direct interconnection, improving network performance and reducing costs.

4.4 Data Collection and Preprocessing

For this application, the needed data consists of date/time stamp, AS path and traffic amount. Data is collected from non-public sources which are considered a corporate internal information. Current data granularity, used in this thesis, is peak data amount per AS number per day, as this is the only data currently which exists for period of 2 years. Future development is planned to include better granularity of data and for longer than 2 years.

5 Feature Engineering

5.1 Identifying Key Features

Identifying the most relevant input features is a critical step in building accurate machine learning models for peering partner prediction. The features selected must reflect the operational and economic drivers behind peering decisions. Within this thesis, the primary focus is placed on traffic throughput, as it directly influences transit costs and indicates whether a private interconnection would provide measurable financial benefit to the ISP.

Latency may also provide strategic value in specific cases, such as when traffic includes real-time applications or latency-sensitive services but, its influence on initial peering decision-making is generally secondary to throughput volume for large-scale ISP interconnection to which latency has an affect specifically when TCP protocol is being used for data transmit.

Packet loss is considered a less relevant feature in this context, as impairment detected within the transit network does not inherently indicate performance issues should a private connection be established. Unless the partners current uplink is congested or error prone, the peering partner's traffic quality is expected to remain consistent regardless of routing path.

In addition to performance-related variables, topological and routing indicators provide value for model constraints. Features such as the AS appearing first in the AS-Path help surface peers currently reachable primarily through transit providers, focusing predictions on relationships with the highest cost-reduction potential.

Finally, location-based features—specifically POP presence and shared Internet Exchange Points—are incorporated to ensure that predicted high-value peers also have practical options for physical interconnection. Location alignment between the ISP and the candidate AS is validated using publicly available information from PeeringDB, ensuring that feasible physical interconnection options exist. (PeeringDB 2025)

By prioritizing features that combine traffic economics, network architecture, and physical feasibility, the system is positioned to recommend peering candidates that offer both technical and business advantage.

5.2 Features relevant to peering decisions (e.g., latency, packet loss, throughput).

Relevant features are dependent to ISP peering policy, most dominant of these features is throughput, though for some use cases latency can be a driving factor.

Packet loss is not very relevant driver for peering selection, one can assume that if an ISP has packet loss in their network, it would be the same if traffic was flowing through a transit provider.

Throughput is a major driver for making peering decisions. It helps bring the ISP transit costs down, as transit providers usually charge a fixed fee based on average usage of the link and a raised fee for any possible overages of the agreed baseline.

5.3 Feature Importance Assessment

For application features, we need to be able to limit the network amount to analyze, one way to do this, is to use a minimum traffic flow amount which must be reached by the other party before we proceed to analyze further. Other factor to consider is the first ASN on the as-path, as primarily the need is to optimize traffic away from transit partners and by analyzing the traffic for ISPs visible through transit provider gives us this.

Other built in application features include comparing Elisa Oyj POP locations against the peering candidate locations by comparing location information from PeeringDB.

Running multiple prediction models is important, as each one of the models has its benefits and drawbacks. Relying on one model could yield misleading results. Features mentioned above are shown on Figure 2.

Figure 22. Presentation of GUI

6 Model Development

6.1 Choosing the Right Model

Selecting a single ML model for the predictions would be unwise as the traffic profiles vary depending on the ISP customer base. For example a large content provider, such as Netflix, has quite established traffic patterns, but ISP hosting a new break through service might show only small indications on their traffic patterns before the service gathers large amount of users. For Netflix, ARIMA model would be a good fit, but for new traffic from ISP, the Prophet model would probably show better fitting predictions.

6.2 Selecting ML models based on data characteristics.

As multiple ML models is used for same data set, selecting the appropriate model or results of a model, is left to the end user, outputs of the models are shown in figure 3. There are variables which can not be quantified to any model, such as customer base changes of an ISP, new services and end user trends which are very difficult to predict.

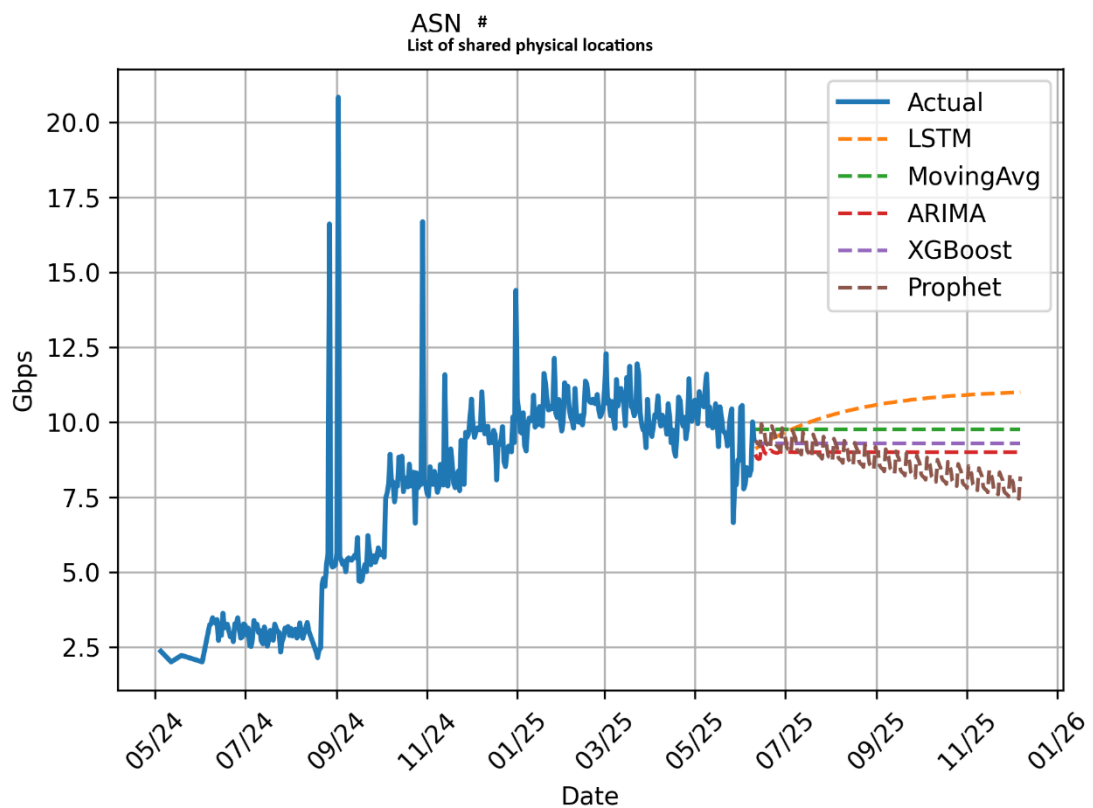


Figure 33. Prediction graph showing different models

7 Evaluation Metrics

7.1 Metrics for Performance Assessment

Performance accuracy can be assessed by comparing predicted data to actualized data, but only retrospectively.

7.2 Define relevant metrics.

Relevant metrics to compare the predictions against are the actualized values within the predicted timeframe, keeping in mind that the trend of the data is more important than the actual values.

7.3 Discuss how to compare ML model performance against traditional methods.

ML model results can and should be compared against the actualized results for few prediction cycles before making decision of taking ML results to production or not. Currently the accuracy is hugely affected by the poor granularity of the data. As we are limited to use only daily values of 95th percentile of traffic throughput, we do not get an idea of the actual data volumes. Current data also has gaps on it, as can be seen from the graph, figure 4, below.

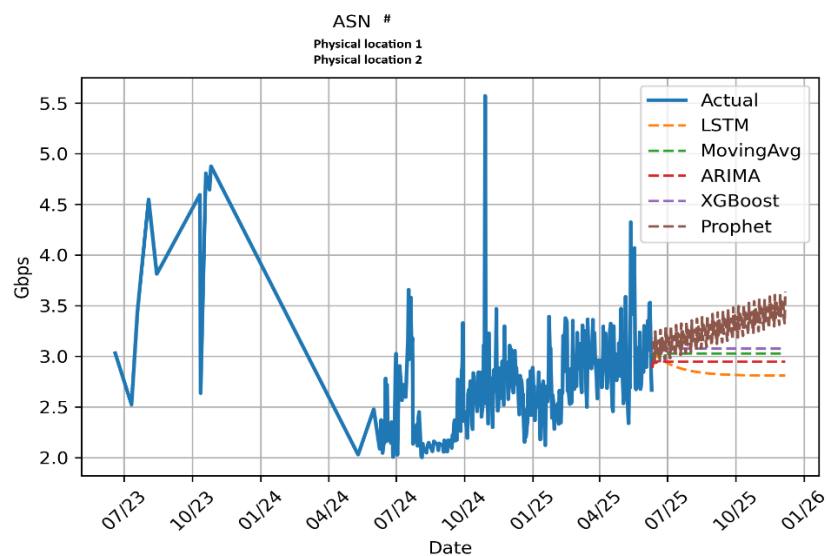


Figure 4. Prediction graph showing missing data points from 07/23 to 06/24

8 Case Studies and Applications

While there are academic papers about ML usage in peering selection, like “Peering Partner Recommendation for ISPs using Machine Learning” (Alam et al. 2025) and “Meta-Peering:

Automating ISP Peering Decision Process” (Uysal et al. 2021), those studies focus on Location, ISP size and other metrics, but they do not utilize any actual traffic data and therefore are lacking in decision making.

8.1 Lessons Learned

As the need is quite niche, there are no publicly available solutions to compare against, so great care needs to be taken before actualizing any actions based on the output of this thesis work.

9 Challenges and Limitations

9.1 Technical Challenges

To limit the possible risks of disrupting any production environment, the thesis work (software) has been done on a separate laptop, not on any server, moving the software to a server environment will cause some modifications to the software and to how the source data is made available for the program.

9.2 Address issues such as model interpretability, scalability, and data availability.

Current setup is enough to produce preliminary results, but for actual production setup we need to migrate the program and data sources to more sustainable environment. Currently the source data is imported to the program from CSV files, which is not ideal but was easily achievable for the proof-of-concept phase. As we need to scale up the source data, a database for it is needed and the data granularity issue needs to be resolved. Currently the most detailed traffic data is available only for the past 3 months, after that the data is sampled down, and we lose many data points.

9.3 Ethical Considerations

As the data itself does not contain any identity data of consumers, there are no real-world ethical constraints.

9.4 Discuss potential biases and ethical implications in ML applications.

Using Machine Learning does not and should not eliminate the need for human oversight, there are many aspects on peering and traffic volume predictions which cannot be purely based on past data.

10 Future Directions

10.1 Trends in ISP Peering

Currently data storage and computing is rapidly moving to large datacenters around the globe (McKinsey & Company 2024), there is a real need for the ISPs to build connectivity to these big service providers, such as Google, Microsoft and Amazon. Automation and machine learning play a vital part in finding and implementing private peering sessions facing ISP's which host new emerging services before the traffic volumes would stand out in traditional evaluation methods.

10.2 Explore future trends and technologies that may impact peering strategies.

As large content providers seek cost savings in their data delivery, they seem to outsource this to smaller content delivery partners, this can lead to rapidly shifting data sources, which can make predictability of any peering setup difficult.

10.3 Further Research Opportunities

Future development could include analyzing the AS paths and ingress locations where the peer AS is seen, this would allow Elisa to ensure redundant routes without relying on the transit providers to carry the traffic should the PNI fail.

10.4 Suggest areas for further exploration and improvement.

Future enhancements to the solution can significantly increase its accuracy, scalability, and operational usefulness. First, improvements in data collection methods and granularity are essential. The current setup relies on CSV files and includes only two years of daily peak traffic samples, with higher-resolution data available for only the most recent months. Populating the database with more granular data will take time, but it could be useful to other projects as well in the future. Transitioning to a centralized database and storing full-granularity traffic measurements would improve forecast quality and allow for more precise model training and evaluation.

Second, the system could be expanded to incorporate additional technical and routing-level metrics, such as ingress location analysis and AS-path diversity. Perhaps even physical routes could be evaluated in this process. This would help ensure that private peering not only reduces transit usage but also enhances redundancy and resilience.

Third, future work should explore adaptive and automated operational workflows, such as integrating the model outputs directly with peering decision platforms or API-driven provisioning systems. This would enable proactive deployment of private network interconnects when traffic growth probabilities surpass operational thresholds.

Finally, incorporating real-time telemetry, continuous learning pipelines, and anomaly detection, such as DDOS attacks, could allow the system to recognize and react to traffic volatility caused by industry trends (e.g., rapid growth from newly emerging services or redistribution of content delivery responsibility).

As telecoms industry shifts toward highly dynamic peering ecosystems, addressing these research areas would substantially increase the system's effectiveness, reduce operational risk, and support more autonomous, data-driven interconnection strategies.

11 Conclusion

11.1 Summary of Findings

The thesis demonstrates that machine learning can successfully leverage historical ISP traffic data to forecast future traffic volumes between an ISP and potential peering partners. Traditional peering-selection relies heavily on manual evaluation, heuristics, and reactive decision-making, which delays cost-saving outcomes. By applying multiple ML time-series models, the implemented tool enables faster, data-driven identification of suitable private peering candidates, especially when growing traffic trends are observed.

11.2 Potential of ML in optimizing peering partner selection.

Machine learning significantly improves peering strategy by:

- 1) Identifying emerging traffic growth patterns before they exceed thresholds
- 2) Allowing ISPs to proactively deploy infrastructure
- 3) Reducing transit-related operational costs
- 4) Improving performance by steering traffic over direct paths

Multiple models (ARIMA, Prophet, LSTM, etc.) allow adaptability to different traffic profiles and provide operational insight that traditional traffic analysis cannot.

11.3 Final Thoughts

Although the solution developed is a proof-of-concept, the results indicate substantial value in incorporating ML into peering evaluations. However, not all variables affecting partner suitability are quantifiable, having face-to-face meetings in peering forums gives more insight to these variables. ML output still requires professional interpretation, we do need to incorporate the knowledge of the ISP future plans and technological development to the decision making. Changes in ISP customer bases and services must remain part of decision-making as it gives insight for possible future emerging services. Human expertise remains

essential, but ML can augment it with stronger predictive intelligence, allowing better time utilization for most interesting peering partner prospects.

11.4 Significance of the research.

This research contributes meaningfully to a niche but high-impact field and highlights how automation and predictive analytics can modernize ISP interconnection strategy. With further development – including improved granularity, scalable deployment, and integration of real-time network metrics – the approach has the potential to become an industry-relevant decision-support system.

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Figures

Figure 1. Peering and Transit

Figure 2. Presentation of GUI

Figure 3. Prediction graph showing different models

Figure 4. Prediction graph showing missing data points from 07/23 to 06/24